
Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Activation steering has recently been shown to reproduce a known protein property (thermostability) in ESM2-650M via difference-of-means vectors. We ask whether a scoring proxy’s correlation with real experimental labels predicts whether steering toward it works, and find that it does not. Under a pre-registered six-criterion protocol, we test five novel target properties and report the failures alongside the successes. Binding affinity’s proxy is the best-validated in our study, with $r = 0.80$ held-out and confirmation from a weight-shuffle null control and held-out-position generalization. Yet it steers nothing. A substantially weaker proxy ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated effect. Immunogenicity is killed at the proxy-validation gate before any model run, after a proxy that looks like a real signal ($r = 0.38$) turns out to be a source-organism confound. Expression yield passes the naive check but fails residue-exclusion; a follow-up diagnostic explains why: its steering vector is a partial geometric echo of a different, already-validated target’s direction. This mirrors an independent, component-level finding in the same model family: attention heads ranked by correlation with real 3D contacts show near-zero relationship to their causal ablation effect. We further show a single-seed result can mislead in criterion-specific ways. We also describe two near-misses caught in our own pipeline: a spurious pass from comparing arms at an unsafe intervention strength and a robustness check that holds on only 2 of 3 seeds. These cases show why this level of scrutiny is necessary before a steering result is trusted. We release the full protocol, all target harnesses, and the seed-replication data.

1 Problem

Task and setting. Difference-of-means activation steering adds a direction to a protein language model’s residual stream. The direction is built from the mean activation difference between high- and low-property example groups. This method has been shown to reproduce known thermostability biology in ESM2-650M [Huang et al., 2025]. The natural next question for anyone applying this technique to a new biological property is whether a scoring proxy’s correlation with real experimental labels predicts whether steering toward it will work.

Assumptions being tested. Adopting this technique for a new target relies on two assumptions. First, validating a compositional proxy against real labels is a reliable gate before a steering experiment. Second, a single steering run that clears a significance threshold is trustworthy evidence of a causal effect. We test five novel target properties: binding affinity, catalytic activity, intrinsic disorder, immunogenicity, and expression yield. All are tested under one pre-registered protocol (§2) designed specifically to stress-test both assumptions, and report every outcome, not only the successes.

Table 1: Summary of all five novel target properties. Proxy r is Pearson correlation against real labels on a held-out split, computed before any steering run.

Target	Dataset	Proxy r	Steering result	Verdict
Binding affinity	ProteinGym DMS (KRAS/DARPin)	0.80	flat null, all α	KILL
Catalytic activity	DLKcat (cross-protein)	0.22	PASS, 3/3 seeds	PASS
Intrinsic disorder	DisProt	0.45	PASS, 3/3 direction; 2/3 robustness	PASS*
Immunogenicity	IEDB (4-tier)	≤ 0.10	not run	KILL (pre-run)
Expression yield	eSol	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1, 2, 4, 6, and 2 of 3 seeds on criterion 3 (§3.1).

Target metrics. For each property, we measure (1) Pearson r of the compositional proxy against real experimental labels on a held-out split, computed before any steering run; (2) the real steering direction’s effect size against a matched-norm random-direction control, via paired bootstrap; (3) whether that effect survives a residue-exclusion robustness check; (4) whether the result replicates across independent random seeds. A “desired improvement” would be a positive, monotonic, seed-stable relationship between proxy validity and steering effect size. We do not find one.

2 Proposed Approach

We lock a six-criterion decision rule before testing any new target and compute it automatically at the end of each run. Criterion 1 requires the real direction to beat a matched-norm random control via paired bootstrap with 10,000 resamples. Criterion 2 requires a monotonic dose-response across at least three strengths within a safe operating range (§4). Criterion 3 requires the effect to survive residue-exclusion, and criterion 4 requires validation of the proxy against real labels before steering. Criterion 5 requires the result to beat the best existing technique when one exists, and criterion 6 requires at least 150 held-out sequences. A target failing any criterion is a KILL; one passing 1–2 but failing 3 or 4 is AMBIGUOUS.

The core mechanism under test in each of the five targets is identical: build a per-layer difference-of-means steering vector from real high/low-property sequence groups, add it to ESM2-650M’s residual stream at generation time, and score the result with a compositional proxy validated in advance. The hypothesis under which this is expected to work is Huang et al.’s finding for thermostability [Huang et al., 2025] generalizing to other properties whenever the scoring proxy is itself well-validated. We also recap an independent, component-level precondition test: Vig et al. [Vig et al., 2021] found an attention head in a protein language model whose attention pattern strongly correlates with real 3D contacts; we ablate it directly and measure the actual causal effect, and extend this to every head in the model (480 total), to test whether correlational salience predicts causal necessity one level below the steering question.

3 Observed Outcome

The central negative result: proxy validity and steering effect size show no positive relationship (Table 1, Figure 1). Binding affinity’s proxy is the best-validated of any target we tried ($r = 0.80$ held-out, confirmed by a weight-shuffle null control giving $r = -0.001$, held-out-position generalization at $r = 0.54$, and mutational-load extrapolation at $r = 0.70$). Despite this validation, the steering effect is a flat, unambiguous null at every tested α , with zero degenerate sequences at any strength. Catalytic activity’s proxy is the weakest of the four runnable targets ($r = 0.22$) and produces a clean, monotonic, residue-robust effect that replicates identically across three independent seeds (Figure 2).

Component-level precondition, same pattern. Vig et al.’s attention head [Vig et al., 2021] shows $12.9\times$ contact enrichment over background, but ablating it produces an effect statistically indistinguishable from ablating a low-correlation control head. A full sweep of all 480 heads, ranked purely by measured causal effect with zero correlational information, shows the correlational top pick

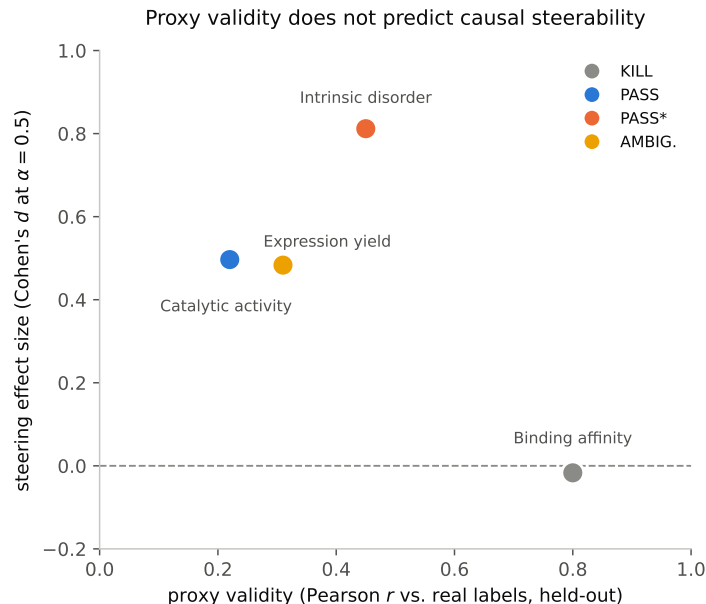


Figure 1: The four runnable targets from Table 1, plotted as proxy validity (Pearson r) against steering effect size (Cohen’s d at $\alpha = 0.5$). The best-validated proxy (binding affinity, $r = 0.80$) produces the smallest effect (indistinguishable from zero); a proxy less than half as strong (catalytic activity, $r = 0.22$) produces one of the two largest effects. There is no visible positive trend.

73 ranks 313th of 480, and 166 of 480 heads (35%) show exactly zero measurable causal effect. This
74 confirms that the result is not due to an unlucky choice of control.

75 **Immunogenicity: killed before any model run.** Compositional proxies that look predictive on a
76 surrogate binding-affinity tier ($r = 0.37$ – 0.43) collapse to near-chance on the real T-cell-response
77 endpoint ($r \leq 0.10$), and a full-length-antigen proxy that appears to reach $r = 0.38$ is a source-
78 organism confound (organism identity alone explains 58% of the apparent signal; the same proxy’s
79 correlation flips to -0.32 once organism is held out of cross-validation). No proxy survives criterion
80 4, so no steering script was written.

81 **Expression yield: passes the naive check, fails the real one.** The proxy validates at $r = 0.31$ –
82 0.34 and produces a significant, dose-responsive steering effect. The effect collapses to statistical
83 insignificance under residue-exclusion, which is the same compositional-collapse pattern our protocol
84 exists to catch.

85 3.1 Intrinsic disorder: a seed-sensitive result, reported honestly

86 Using DisProt, we build a proxy from the published TOP-IDP disorder scale [Campen et al., 2008]
87 ($r = 0.45$ held-out, the strongest proxy in our sweep, confirmed at the per-residue level with AUC
88 0.71). The steering effect is significant and monotonically dose-responsive on every one of three
89 independent seeds (Figure 2). However, the residue-exclusion robustness check specifically passes
90 on 2 of 3 seeds (36–42% of effect magnitude retained, CI excludes zero) and fails on the third (10%
91 retained, CI crosses zero). The effect’s existence and direction are seed-stable, but the artifact check
92 that rules out pure compositional collapse is not uniformly stable (Figure 3).

93 4 Reason for Failure

94 **Binding affinity’s null is mechanistically explained, not just observed.** The vector-building
95 groups for this target are single-backbone point mutants. They differ by 1–2 residues from a shared
96 188-residue reference, which gives the difference-of-means construction very little raw compositional

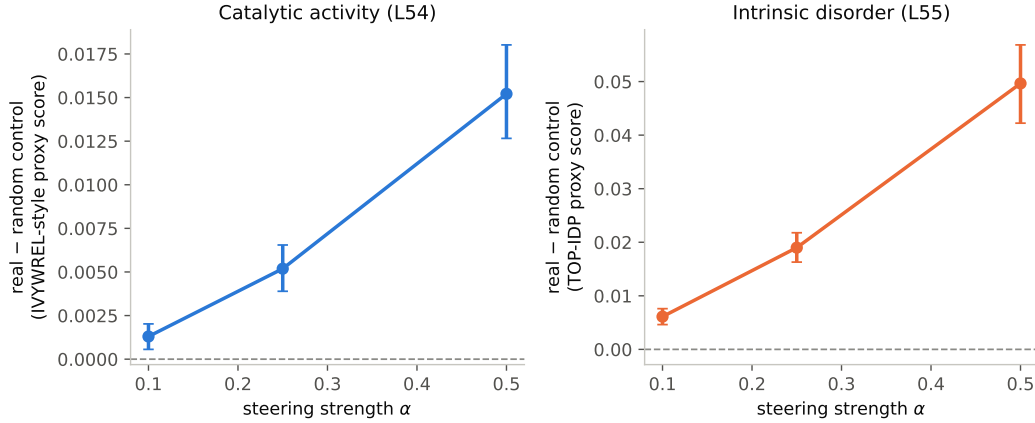


Figure 2: Dose-response for the two targets that pass criterion 2. Both show a clean, monotonic increase with 95% bootstrap CIs bounded away from zero at every point.

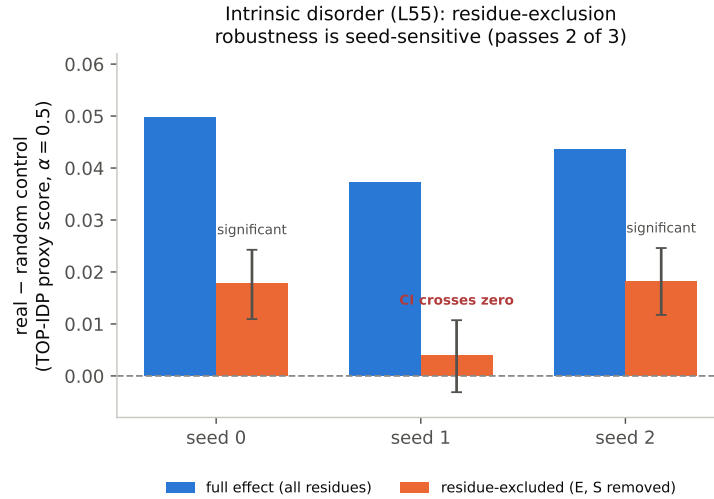


Figure 3: The intrinsic-disorder target’s effect (blue) and the same effect after the two dominant substituted residues are excluded (orange, 95% bootstrap CIs), across three independent seeds. The full effect is stable across all three; the excluded-residue effect is significant on seeds 0 and 2 but its CI crosses zero on seed 1. The underlying effect is unchanged, but the criterion produces three different verdicts.

97 signal to use. This limitation is independent of whether binding affinity is causally steerable. We
 98 measure it directly: mean per-residue compositional distance between the low/high vector-building
 99 groups is 0.003 for this target, versus 0.02–0.05 for the three cross-protein datasets in our sweep. A
 100 well-validated scoring function does not guarantee the underlying difference-of-means construction
 101 has anything to act on.

102 **Expression yield’s failure is explained by vector geometry, not discarded as noise.** We compute
 103 cosine similarity between this target’s steering vector and every other target’s, layer by layer, and find
 104 substantial overlap with the intrinsic-disorder target’s vector: +0.38 on the full 33-layer concatenation,
 105 rising to +0.56–0.67 at the three deepest layers. This is consistent with the independently documented
 106 biology (disordered regions reduce solubility and expression yield). Expression yield’s apparent
 107 signal is best understood as a partial, noisier echo of disorder’s already-validated real direction,
 108 filtered through a proxy more prone to compositional collapse than TOP-IDP. It is not independent
 109 evidence of a sixth steerable property.

A near-miss that would have hidden the real boundary condition. As a mechanistic follow-up on thermostability, we tested whether restricting steering to the 5 layers previously identified as causally necessary preserves the effect of steering all 33. A first implementation selected its best-performing α from the full tested range, including $\alpha \geq 1.0$, and reported a clean pass: the 5-layer subset appeared to outperform the full intervention at $\alpha = 2.0$. This was an artifact of comparison: at that α , the 33-layer intervention had already fully collapsed into degenerate output (zero non-degenerate evaluation sequences), while the 5-layer subset had not yet collapsed. It “won” only because it broke down later, not because it steered harder in any comparable regime. Restricting the comparison to a pre-established safe operating range and rerunning end to end gives the real result: the 5-layer subset produces a real, residue-robust effect, but retains only 43–59% of the full effect size. This is informative but not the unqualified success suggested by the uncollapsed comparison.

Boundary conditions and actionable takeaways. Proxy validation strength is not a reliable filter for which targets are worth attempting. Check the raw compositional separation between vector-building groups, not just the scoring function’s correlation with ground truth. Any α -selection or best-config comparison must be restricted to a range where every compared arm is known to be non-degenerate, or a collapse-order artifact can produce a spurious win. A criterion-specific robustness check, such as residue-exclusion, can have different seed sensitivity than the headline effect and therefore needs its own multi-seed check. An AMBIGUOUS or unexplained result also warrants a cheap vector-geometry cross-check against other validated targets in the same study before being dismissed as noise.

5 Related Work

Huang et al. [2025] introduce difference-of-means activation steering for protein language models and validate it on thermostability; we extend their harness to five new properties under a pre-registered protocol and test, for the first time to our knowledge, whether proxy correlation strength predicts steering success across multiple unrelated properties. Vig et al. [2021] identify the attention head we redo as a genuine causal test and extend to a full 480-head sweep. Adams et al. [2025] train sparse autoencoders on ESM-2 and show via the correlational method of linear probing that known determinants of thermostability and localization are identifiable in SAE feature space. Their one steering experiment is a coherence check, not a validation against a real label, and their own discussion states that “investigating how to steer the model in useful ways remains an open direction for future work.” We answer a version of that question for a non-SAE steering mechanism, specifically for whether correlational validity predicts causal usefulness.

6 Limitations

All experiments use a single model, ESM2-650M, at a single scale. All proxies are compositional heuristics chosen for interpretability and speed, not learned predictors. A stronger, non-compositional proxy might behave differently. Each target’s final proxy was selected as the best of several candidates, typically 7 to 9, scored on the same held-out split whose r is reported as the validation number. We did not use a separate train/select/test split. This garden-of-forking-paths bias optimistically inflates every reported proxy- r , so the absolute values in Table 1 should be read as upper bounds. It does not obviously overturn the central ordinal comparison because binding affinity remains the best-validated target and remains null.

We ran many significance tests overall: 5 targets across 5 alphas, 3-seed reruns for two targets, and the 480-head sweep. We did not apply a multiple-comparisons correction. Requiring multi-alpha and multi-seed agreement for any PASS partially mitigates this issue, but borderline criteria remain close to the noise floor. One example is the residue-exclusion check in §3.1, which is significant on only 2 of 3 seeds. We tested seed robustness for catalytic activity and intrinsic disorder, but not all five targets. We also did not run structural or functional plausibility checks, such as predicted-fold confidence, on generated sequences. A PASS therefore does not rule out otherwise implausible proteins.

Every steering result here is causal about the model’s activation space. It reflects a real intervention evaluated against a random-direction control and a residue-exclusion check, but it is not validated

161 against a real biological assay. A PASS means “causally steers a validated compositional proxy,” not
162 “causally engineers the underlying biological property.”

163 **LLM usage disclosure**

164 Large language model assistance (Claude, Anthropic) was used throughout this project for code
165 authoring, experiment scripting, literature lookup, and drafting of this manuscript. All experimental
166 design decisions, the pre-registered protocol, and the interpretation of results were made and verified
167 by the authors; every quantitative claim in this paper is backed by a runnable script and a committed
168 results file, checked against the paper’s prose during preparation.

169 **Code and data availability**

170 Code and results supporting every number in this paper are publicly available in the project repository,
171 which is released regardless of submission outcome.

172 **Reproducibility statement**

173 All five target properties in Table 1 reuse one fixed harness (steering-vector construction, matched-
174 norm random control, paired bootstrap significance test, degeneracy filter), varying only the target
175 dataset and scoring proxy, so a single implementation bug cannot selectively explain a subset of the
176 reported results. Every quantitative claim, including both near-misses in §4, is backed by a committed,
177 re-runnable script and a saved results file; the two multi-seed claims (catalytic activity, intrinsic
178 disorder) rerun the identical pipeline end to end per seed rather than resampling a single run’s output.

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